

Generation Interconnection System Impact Study Report

For Queue Project Q18-0966

Sheridan Alte 138 kV

January 29, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0966. The Interconnection Customer proposes a 148.4 MW solar photovoltaic facility interconnecting at the Sheridan Alte 138 kV substation in the Virginia area. The request was received on March 19, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 435 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 14 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$16,063,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of Moonshine / Hondo 138 KV, the Beardsley St / Neosho 138 KV loads to 159.5% of its 204 MVA rating. Project DFAX: 11.4%.

For the loss of Vc Porterdale Hydroelectric / Bonsack 138 KV Line, the Vinton / Hammond 138 KV loads to 132.4% of its 184 MVA rating. Project DFAX: 20.4%.

Vc Porterdale Hydroelectric / Vinton 138 KV: 129.4% loading (196 MVA) following the outage of the Bonsack / Vc Porterdale Hydroelectric 138 KV; DFAX = 20.4%.

President Park / Hammond 138 KV: 128.6% loading (190 MVA) following the outage of the Vc Porterdale Hydroelectric / Bonsack 138 KV; DFAX = 20.4%.

For the loss of Hondo / Moonshine 138 KV, the Hancock / Ash St 138 KV loads to 122.6% of its 232 MVA rating. Project DFAX: 11.4%.

The President Park Substation / Roanoke 138 KV Line is overloaded to 120.8% of its 184 MVA emergency rating for the loss of the Vc Porterdale Hydroelectric / Bonsack 138 KV. The project contributes 20.4% DFAX (30.3 MW) to this overload.

For the loss of Riverview / Person Substation 345 KV Line, the Clover / Leesville 345 KV loads to 115.1% of its 1,241 MVA rating. Project DFAX: 23.2%.

Huntington Court / Bonsack 138 KV line: 108.6% loading (282 MVA) following the outage of the Vc Porterdale Hydroelectric / Vinton 138 KV; DFAX = 27.3%.

Vc Porterdale Hydroelectric / Bonsack 138 KV: 108.2% loading (310 MVA) following the outage of the Vc Porterdale Hydroelectric / Vinton 138 KV Line; DFAX = 27.3%.

Salm Electric Department / Neosho 138 KV: 106.5% loading (165 MVA) following the outage of the Moonshine / Hondo 138 KV L; DFAX = 5.7%.

For the loss of Nacoochee / S Texa 138 KV line, the South Crk / Wythe 138 KV L loads to 105.7% of its 517 MVA rating. Project DFAX: 5.6%.

Bear / Leavitt 345 KV Cir. 2: 103.6% loading (1,957 MVA) following the outage of the Bear / Leavitt 345 KV #1; DFAX = 14.9%.

The Mecklenburg Power Sta / Person 345 KV is overloaded to 102.8% of its 2,711 MVA emergency rating for the loss of the Bear / Clover 345 KV. The project contributes 27.8% DFAX (41.3 MW) to this overload.

Hondo / Moonshine 138 KV Line: 102.6% loading (213 MVA) following the outage of the Beardsley St / Neosho 138 KV Line; DFAX = 15.6%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Beardsley St / Neosho 138 KV	Rebuild line with higher-capacity conductor	\$4,906,000	32% (\$1,561,000)
Vinton / Hammond 138 KV	Rebuild line with higher-capacity conductor	\$3,375,000	15% (\$512,000)
Vc Porterdale Hydroelectric / Vinton 138 KV	Rebuild line with higher-capacity conductor	\$2,983,000	15% (\$452,000)
President Park / Hammond 138 KV	Rebuild line with higher-capacity conductor	\$2,391,000	15% (\$363,000)
Hancock / Ash St 138 KV	Rebuild line with higher-capacity conductor	\$2,733,000	32% (\$870,000)
President Park Substation / Roanoke 138 KV Line	Rebuild line with higher-capacity conductor	\$3,045,000	15% (\$462,000)
Clover / Leesville 345 KV	Rebuild line with higher-capacity conductor	\$163,181,000	5% (\$7,954,000)
Huntington Court / Bonsack 138 KV Line	Reconductor line	\$674,000	14% (\$96,000)
Vc Porterdale Hydroelectric / Bonsack 138 KV	Reconductor line	\$926,000	14% (\$131,000)
Neosho / Salem Electric Department 138 KV	Reconductor line	\$291,000	100% (\$291,000)
South Crk / Wythe 138 KV L	Reconductor line	\$3,535,000	3% (\$96,000)
Bear / Leavitt 345 KV Cir. 2	Reconductor line	\$33,131,000	3% (\$1,097,000)
Mecklenburg Power Station / Person 345 KV	Reconductor line	\$27,011,000	4% (\$1,120,000)
Hondo / Moonshine 138 KV Line	Reconductor line	\$1,998,000	53% (\$1,058,000)
	Total allocated to project		\$16,063,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.